

Class 8 — Set A

Mateng EduFest 2026

1. Which of the following is not true? Rational numbers are:

- A) Closed under addition.
- B) Closed under subtraction.
- C) Closed under multiplication.
- D) Closed under division.

2. The reciprocal of 0 is

- A) 1
- B) -1
- C) 0
- D) Not defined

3. Observe the histogram given. The number of girls having height 145 cm and above is

- A) 5
- B) 10
- C) 17
- D) 19

4. Ram put some buttons on the table: 4 blue, 7 red, 3 black and 6 white in all. All of a sudden, a cat jumped on the table and knocked one button onto the floor. What is the probability that the button on the floor is not blue?

- A) $\frac{7}{20}$
- B) $\frac{3}{5}$
- C) $\frac{1}{4}$
- D) $\frac{4}{5}$

5. If you spin a spinner divided into three unequal sections — one large blue, one medium red, and one tiny green — are the outcomes of landing on a specific colour equally likely?

- A) Yes, because there are only three colours.
- B) Yes, because each colour is a possible outcome.
- C) No, because the probability of landing on each colour depends on the size of the section.
- D) No, because the green section is impossible to land on.

6. If one member of a Pythagorean triplet is $2m$, then the other two members are

- A) $m, m^2 + 1$
- B) $m^2 + 1, m^2 - 1$
- C) $m^2, m^2 - 1$
- D) $m^2, m + 1$

7. Given that $\sqrt{4096} = 64$, the value of $\sqrt{4096} + \sqrt{40.96}$ is

- A) 74
- B) 60.4
- C) 64.4
- D) 70.4

8. A 5 m long ladder is leaned against a wall. The ladder reaches the wall at a height of 4 m. Find the distance between the wall and the foot of the ladder.

- A) 9 m
- B) 1 m
- C) 3 m
- D) 20 m

9. What is the smallest square number that is divisible by each of the numbers 4, 9 and 10?

- A) 360
- B) 900
- C) 3600
- D) 32400

10. There are $2n$ non-perfect-square numbers between the squares of the numbers n and $n + 1$. If there are 112 non-square numbers between two consecutive perfect squares, what is the smaller of the two consecutive base numbers?

- A) 56
- B) 57
- C) 111
- D) 112

11. If an odd number squared is expressed as the sum $180 + 181 = 361$, what is the original odd number?

- A) 17
- B) 19
- C) 21
- D) 361

12. Which of the following correctly describes the relationship between zeros in a base number and zeros in its perfect cube?

- A)** A perfect cube always has an even number of zeros.
- B)** A perfect cube cannot end with exactly two zeros.
- C)** The number of zeros in the cube is half the number of zeros in the original number.
- D)** A perfect cube always ends in exactly three zeros.

13. A student wants to know if 69 is a perfect square using the repeated-subtraction method. What will be the outcome of successively subtracting odd natural numbers starting from 1?

- A)** The subtraction will reach 0 after 8 steps, proving it is a square.
- B)** The subtraction will reach 0 after 9 steps, proving it is a square.
- C)** The subtraction will result in a negative number without ever reaching 0, proving it is not a perfect square.
- D)** The subtraction will reach exactly 1, proving it is not a perfect square.

14. Which number is the Hardy–Ramanujan number?

- A)** 1000
- B)** 1225
- C)** 1729
- D)** 2178

15. Which of the following statements comparing perfect squares and perfect cubes is true?

- A)** Both a perfect square and a perfect cube can end with exactly two zeros.
- B)** The number of zeros at the end of a perfect square must be a multiple of 2, while for a perfect cube it must be a multiple of 3.
- C)** A perfect square can end in the digit 8, but a perfect cube cannot.
- D)** Odd numbers have even squares, but odd cubes.

16. Two terms are considered "like terms" if:

- A)** They have the same coefficient.
- B)** They have the same variables with the same exponents.
- C)** They are both monomials.

D) They have the same number of factors.

17. Subtract $(3p^2q - 3pq^2)$ from $(2p^2q + 2pq^2)$.

The result is:

- A)** $5p^2q - pq^2$
- B)** $-p^2q + 5pq^2$
- C)** $p^2q - 5pq^2$
- D)** $-p^2q - pq^2$

18. Which of the following properties is true for ALL parallelograms?

- A)** Diagonals are equal in length.
- B)** Adjacent angles are complementary.
- C)** Diagonals bisect each other.
- D)** All sides are of equal length.

19. In a regular polygon, if each exterior angle measures 24° , how many sides does the polygon have?

- A)** 12
- B)** 15
- C)** 18
- D)** 20

20. Why is a rectangle considered a convex quadrilateral?

- A)** Because its diagonals are equal in length.
- B)** Because all its interior angles are right angles.
- C)** Because all line segments joining any two interior points lie wholly within the interior.
- D)** Because its diagonals bisect each other.

21. Meenu bought two fans for ₹1200 each. She sold one at a loss of 5% and the other at a profit of 10%. Then she made

- A)** ₹60 gain
- B)** ₹60 loss
- C)** ₹180 gain
- D)** ₹180 loss

22. An article was purchased for ₹1239, including GST of 18%. Find the price of the article before GST was added.

- A)** ₹1221
- B)** ₹1050
- C)** ₹1511
- D)** None of the above

23. A person invests ₹8,000 at a variable rate of 4% for the first year and 5% for the second year. Calculate the difference between the compound interest and the simple interest, if simple interest were calculated at a flat 4.5% per annum for 2 years.

- A) ₹4
- B) ₹8
- C) ₹12
- D) ₹16

24. The following pie chart gives the distribution of constituents in the human body. The central angle of the sector showing the distribution of water is

- A) 108°
- B) 70°
- C) 30°
- D) 252°

25. Which of the following is incorrect?

- A) Every perfect square number has two square roots.
- B) Negative prime numbers exist.
- C) There is no number whose square ends with 2.
- D) A divisor can never be zero in a division.

26. Which one is the cheapest item among the following?

- A) MRP ₹1000 with a discount of 20%.
- B) Buy 1 @ ₹1000 and get another for free.
- C) MRP ₹900 and fixed price.
- D) Buy 3 items and pay for only 2, where the MRP of each item is ₹500.

27. Which one of the following is true?

- A) Simple interest is more than compound interest for 1 year for the same principal at the same rate.
- B) Simple interest is less than compound interest for 1 year for the same principal at the same rate.
- C) Simple interest is the same as compound interest for 1 year for the same principal at the same rate.
- D) None of the above.

28. From a rope 40 metres long, pieces of equal size are cut. If the length of one piece is $\frac{10}{3}$ metre, then the number of pieces is

- A) 12
- B) 4
- C) $\frac{1}{12}$
- D) $\frac{400}{3}$

29. You sell 16 articles for the same money as you paid for 20. What is your gain per cent?

- A) 80%
- B) 25%
- C) 20%
- D) 125%

30. The salary of a worker is first increased by 10% and thereafter reduced by 10%. What was the overall change in the salary?

- A) 100% increase
- B) 10% increase
- C) 1% reduction
- D) No change at all

31. A steamer goes downstream from one point to another in 7 hours. It covers the same distance upstream in 8 hours. If the speed of the stream is 2 km/hr, then the distance between the two points is

- A) 224 km
- B) 30 km
- C) 16 km
- D) 14 km

32. $2x + 5x + 7 - 2$ has

- A) 4 terms
- B) No terms at all.
- C) Only one term.
- D) Two terms.

33. For what value of x is the perimeter of the given shape 77 cm?

- A) 11
- B) 10
- C) 9
- D) 12

34. On dividing ₹200 between A and B such that twice A's share is less than three times B's share by ₹200, B's share is

- A) ₹140
- B) ₹80
- C) ₹120
- D) ₹50

35. $(a + b)^2 - (a - b)^2 = ?$

- A) $2a$
- B) $4a$
- C) $4ab$
- D) $2b$

36. The irreducible factorization of $3a^3 + 6a$ is

- A) $3a(a^2 + 2)$
- B) $3(a^3 + 2)$
- C) $a(3a^2 + 6)$
- D) $3 \times a \times a \times a + 2 \times 3 \times a$

37. If $x + 1/x = 5$, then the value of $(x^4 + 1/x^4)$ is

- A) 25
- B) 23
- C) 527
- D) 623

38. The area of a circle is given by the expression $\pi x^2 + 6\pi x + 9\pi$. The radius of the circle is

- A) $x + 3$
- B) 9π
- C) $6\pi x + 9\pi$
- D) Cannot determine

39. The height of a rectangle in a histogram shows the

- A) Width of the class
- B) Upper limit of the class
- C) Lower limit of the class
- D) Frequency of the class

40. Which of the following is not a random experiment?

- A) Tossing a coin
- B) Rolling a dice
- C) Choosing a card from a deck of 52 cards
- D) Throwing a stone from the roof of a building

41. The marks scored by 10 students in an exam are: 30, 20, 40, 30, 60, 80, 90, 80, 40, 100. If the pass mark is 50, then the mean mark of the passed students is

- A) 82
- B) 57
- C) 50
- D) 60

42. A number of the form p/q is said to be a rational number if

- A) p and q are integers.
- B) p and q are integers and $q \neq 0$.

C) p and q are integers and $p \neq 0$.

D) p and q are integers, $p \neq 0$ and $q \neq 0$.

43. Which of the following statements is always true?

- A) $(x - y)/2$ is a rational number between x and y .
- B) $(x + y)/2$ is a rational number between x and y .
- C) $(x \times y)/2$ is a rational number between x and y .
- D) $(x \div y)/2$ is a rational number between x and y .

44. The ratio of the volume of a cube of side 10 cm to a jar of 5 litres is

- A) 2 : 1
- B) 200 : 1
- C) 1 : 5
- D) Cannot determine, as the quantities are not comparable.

45. In a parallelogram PQRS, the bisectors of $\angle P$ and $\angle Q$ meet at O. Then $\angle POQ$ is

- A) An acute angle
- B) A right angle
- C) An obtuse angle
- D) Cannot determine

Questions 46–50

Questions 46–50: There are two or more correct options.

46. Choose the correct statements:

- A) Any improper fraction is a rational number.
- B) There are a finite number of rational numbers between two rational numbers.
- C) 1, 2, 3, 4, 5 are rational numbers between -1 and $11/2$.
- D) A rational number is represented by more than one point on a number line.

47. The pie chart gives the marks scored in an examination by a student in different subjects. If the total marks obtained were 540:

- A) The student scored 105 marks in Hindi.
- B) The student's Maths mark is 25% of his total marks.
- C) The student is getting 80% of his score in Science.
- D) The student gets 30 more marks in Mathematics than in Hindi.

48. Which of the following statements are true?

- A) Very small numbers can be expressed in standard form using positive exponents.
- B) $(-10) \times (-10) \times (-10) \times (-10) = 10^{-4}$
- C) The expression for 4^{-3} as a power with base 2 is 2^{-6}
- D) $0^0 = 1$

49. Which is true regarding $(x^2 - y^2)$?

- A) $x + y$ is a factor.
- B) $x - y$ is also a factor.
- C) $(x + y)(x - y)$ is also a factor.
- D) It divides $(x^4 - y^4)$.

50. Mark the correct options:

- A) You earn ₹10 more when you lend ₹1000 @ 10% per year for 2 years through compound interest instead of simple interest.
- B) You sell an item with MRP ₹100 for ₹80. If you bought the item for ₹50, then your customer is getting a discount of 20% and you are also making a profit of 60%.
- C) If the ratio of boys to girls in a class is 4 : 1, then the number of boys exceeds the number of girls by 60% of the total students.
- D) If you borrow an amount at compound interest and lend the same amount at simple interest at the same rate, you will be rich one day.